Econometrics II (JEB110)

Home Assignment 3

November 18, 2025

Deadline: December 2, 2025, 11:59 PM

Dear students,

You are encouraged to work in groups of two people. In that case, hand only one solution with the names of both participants on the first page. In case you cannot find any colleague to form a group with, you can write an email to Veronika Placha (*85675870@fsv.cuni.cz*) and she will match you with somebody. Submit your solution electronically via SIS. Pack all electronically submitted files into a zipped folder named **"Surname1_Surname2_HA1"**.

The solution of the first problem should be handwritten and might be submitted on paper to the mailbox No.14 (Barbara Pertold-Gebicka) or scanned and submitted electronically. The solution of the second problem can be written in any text-editing software of your choice (e.g., MS Word, LATEX) and should be submitted electronically. Attach both R "log" file with results and "rmd" file with the programming code to your assignment. If using another statistical software, attach the corresponding file(s). Use comments in your programming code to describe in few words what each command means. In case of problems with R, consult the official online manual and help module.

Keep in mind that not only the correctness of the results is assessed, but also the text editing quality is taken into consideration. Answer only what you are asked for.

Note that the use of **Artificial Intelligence (AI)** tools is only allowed to improve your writing style and grammar or to generate inspiration. However, treat the AI tools' output carefully and with caution, respecting the principles of academic integrity and the requirement to produce your own original work that is based on reliable and verifiable sources of information. Simple copying of the text generated by the AI tools is not permitted. Consult the full text of AI guidelines here.

Problem 1

Remember to submit hand-written solution! Consider the following three-equation model:

$$Y_{1t} = \alpha_0 + \alpha_1 Y_{2t} + \alpha_2 X_{1t} + \alpha_3 X_{2t} + u_{1t}$$

$$Y_{2t} = \beta_0 + \beta_1 Y_{3t} + \beta_2 X_{1t} + \beta_3 X_{3t} + u_{2t}$$

$$Y_{3t} = \gamma_0 + \gamma_1 Y_{1t} + \gamma_2 X_{2t} + \gamma_3 X_{3t} + u_{3t}$$

where the $Y's$ are the endogenous variables, the $X's$ the exogenous variables, and the $u's$ the stochastic errors.

(a) Derive the reduced form regression for Y_{2t}.

(b) Determine which of the equations in the above system is identified. Explain.

(c) For the identified equation(s), which method of estimation would you use and why? Explain step by step.

Problem 2

In this exercise, you will analyze a simulated crosssectional dataset that can be found in SIS under the name HA3_2025.xls. The dataset includes the following variables:

- x1 - independent variable 1
- x2 - independent variable 2
- x3 - independent variable 3
- y1 - dependent variable 1
- y2 - dependent variable 2
- x4 - potential proxy variable, highly correlated with x2
- z3 - potential instrumental variable, partially correlated with x3

The dependent variables were generated by the following data-generating process:

$$y = \beta_0 + \beta_1 \cdot x1 + \beta_2 \cdot x2 + \beta_3 \cdot x3 + u, \tag{1}$$

with $\beta_0 = 1$, $\beta_1 = 0.5$, $\beta_2 = 1$, and $\beta_3 = 3$. The random disturbance has mean 0 and standard deviation 1.5 (for $y1$) or standard deviation 4 (for $y2$). In other words, the only difference between $y1$ and $y2$ is the variance of the idiosyncratic part.

PART 1

(a) Inspect the baseline data, i.e., the independent and dependent variables. You might produce a table of descriptive statistics, compute correlations, or draw graphs. Which important conclusions do you derive from inspecting the data?

(b) Use OLS to estimate the following regression model for both $y1$ and $y2$:
$y_i = \beta_0 + \beta_1 \cdot x1_i + \beta_2 \cdot x2_i + \beta_3 \cdot x3_i + u_i$.
Present estimation results in a comprehensive table. Why does the R-squared differ between the two models?

(c) Consider the hypothesis that $\beta_3 = 3$. Is it correct, given the data-generating process? Test this hypothesis using the estimates obtained in (b). Comment.

(d) Imagine that variable $x1$ is unobservable. Estimate the following model by OLS:
$y_i = \beta_0 + \beta_2 \cdot x2_i + \beta_3 \cdot x3_i + e_i$.
Test the hypothesis that $\beta_3 = 3$. Does the test lead to the same conclusion as in (c)? Is the precision of the test the same? Comment.

(e) Now, imagine that variable $x2$ is unobservable. Estimate the following model by OLS:
$y_i = \beta_0 + \beta_1 \cdot x1_i + \beta_3 \cdot x3_i + e_i$.
Test the hypothesis that $\beta_3 = 3$. Does the test lead to the same conclusion as in (c)? Is the precision of the test the same? Comment.

PART 2

Now, delete the variable $x2$ from the dataset. Imagine that the data at your disposal is a real-life dataset and your task is to estimate the relationship between $x3$ and y. You know the general shape of the data-generating process 1, but you do not know the values of the coefficients. Present your work in the form of a mini-report that summarizes your empirical approach, discusses alternative results, comments, and concludes. Attach both the R "log" files with results and ".R" file with the programming commands or an **R Markdown file**. If you use another statistical software, attach the relevant files. Treat the points below as a suggestive checklist.

1. Explain your goal and build a baseline linear regression model that may be used to estimate the relationship between $x3$ and y.

2. Discuss whether the CLM assumptions are satisfied. Are you able to test it? What about the exogeneity assumption (the zero conditional mean)?

3. If any of the assumptions necessary for consistency is not satisfied, explain how will it affect the OLS estimator $\hat{\beta}_3$, i.e., do you expect $\hat{\beta}_3$ to over-, under-, or exactly estimate the true value of β_3? Why?

4. Propose two alternative methods to consistently estimate β_3. Discuss whether your data allows you to apply these methods. Can you verify the assumptions that are necessary for these methods to produce consistent estimates?

5. Run the two above-proposed estimations (or just one, if the other one does not seem trustworthy). Present results in a comprehensive table. You may also want to include OLS estimates for comparison. Do it separately for $y1$ and separately for $y2$.

6. Use the Hausman test to compare the alternative method(s) with OLS, again separately for $y1$ and separately for $y2$. Does the conclusion differ for $y1$ and for $y2$. Why?